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(71) Name of Applicant(s):
1. Srm Institute Of Science And Technology, Ramapuram Campus,
2. Easwari Engineering College,

(72) Name of Inventor(s):
1. Dr Padmavathi B
2. Ricardo Lopez
3. Velpuri Sai Pujitha
4. Jyothirmayee M
5. Nd Tendulkar Jay
6. Yashmitha Sai
7. Sreeshanth

(57) Abstract:

Abstract: The project creates a complete AI system which uses performance optimization methods and automated control techniques to improve Hall-effect thruster (HET) operational efficiency in space propulsion systems. The system operates through a laboratory-scale HET testing facility which uses low-power equipment to test HET systems while plasma actuators work with multiple sensors that track essential live operational data which includes discharge voltage and current and magnetic field strength and vacuum pressure and plume movement. The system uses a complete digital twin model which constantly replicates thruster operations to determine how different operational settings will impact subsequent outcomes. The system uses an advanced AI controller which combines reinforcement learning algorithms with Deep Deterministic Policy Gradient (DDPG) and Proximal Policy Optimization (PPO) methods and Bayesian optimization to control plasma non-linear dynamics. The system functions through an advanced closed-loop system which uses sensors to acquire telemetry data. The AI controller provides the system with recommended optimal set points while actuators which include DC coils and proportional valves control the thruster settings. To maintain safe and dependable operations during high- voltage testing the system uses Kalman filters to process data noise while simulation- first testing prevents the AI from generating dangerous commands. The system uses an autonomous data-driven tuning environment which allows for faster development of satellite propulsion systems and requires less testing work and results in lower development costs.

ADAPTIVE AI HET (HALL EFFECT THRUSTER) TESTBED

Abstract:

The project creates a complete AI system which uses performance optimization methods and automated control techniques to improve Hall-effect thruster (HET) operational efficiency in space propulsion systems. The system operates through a laboratory-scale HET testing facility which uses low-power equipment to test HET systems while plasma actuators work with multiple sensors that track essential live operational data which includes discharge voltage and current and magnetic field strength and vacuum pressure and plume movement. The system uses a complete digital twin model which constantly replicates thruster operations to determine how different operational settings will impact subsequent outcomes. The system uses an advanced AI controller which combines reinforcement learning algorithms with Deep Deterministic Policy Gradient (DDPG) and Proximal Policy Optimization (PPO) methods and Bayesian optimization to control plasma non-linear dynamics.

The system functions through an advanced closed-loop system which uses sensors to acquire telemetry data. The AI controller provides the system with recommended optimal set points while actuators which include DC coils and proportional valves control the thruster settings. To maintain safe and dependable operations during high-voltage testing the system uses Kalman filters to process data noise while simulation-first testing prevents the AI from generating dangerous commands. The system uses an autonomous data-driven tuning environment which allows for faster development of satellite propulsion systems and requires less testing work and results in lower development costs.

Description:

Background of the Invention:

The current process for establishing a space thruster functions as an operation that requires manual engine tuning during engine operation. The procedure requires extended time periods and significant financial resources and it presents a risk of hardware damage because plasma behavior exhibits extreme unpredictability. Most systems only react after something goes wrong. The existing system needs development because it should possess the ability to anticipate upcoming heat and stability problems which it should resolve before they occur.

Summary of the Invention:

The system uses intelligent data analysis to control and improve Hall-Effect Thrusters (HET) operations without requiring ongoing manual adjustments. The system functions as a "brain" for the thruster by combining a digital twin with an artificial intelligence controller to manage intricate plasma dynamics. The system achieves this through a specific technical setup: The AI employs reinforcement learning to analyze thruster data and determine optimal operational settings which enhance both efficiency and stability according to predictive optimization.

The system employs Kalman filters and Bayesian optimization to maintain hardware protection during high-voltage operations by eliminating data interference which would lead to unsafe AI recommendations. The system operates a closed-loop process which enables sensors to monitor the thruster while the AI provides operational recommendations and actuators (including valves and coils) perform immediate control adjustments.

Digital Twin Integration: The virtual model of the thruster operates in parallel with the actual equipment by showing design behavior and forecasting various performance outcomes which testing will implement after their complete development.

Dual-Interface Monitoring: The system operates like modern EV battery management systems because it enables engineers to track thruster performance through a local display and mobile application which synchronizes data between these two systems. The invention enables faster and cheaper space propulsion development because it uses autonomous testing instead of time-consuming manual testing processes.

Core Architectural Layers Power & Logic Stage: High-voltage (HV) power supplies the plasma system while the buck converter produces a 5V output which serves as the microcontroller power source. Multi-Sensor Array: This layer captures live "vitals," which include discharge voltage and current and magnetic field intensity detected by Hall sensors and vacuum pressure and plume light behavior detected by photodiodes.

AI & Simulation Layer:

The AI controller employs reinforcement learning together with Bayesian optimization methods to identify optimal performance set-point measurements.

Digital Twin: A high-fidelity software model that predicts how changes will affect the thruster before they are applied.

The Actuation Interface : connects physical devices through DC coils and proportional valves which receive AI signals to perform immediate magnetic field changes and fuel flow modifications.

Functional Block Diagram Flow The system operates in a continuous cycle to ensure the thruster stays at peak efficiency :

Data Capture: Sensors monitor the current state of the plasma and hardware.
The AI controller uses machine learning to analyze data and propose optimized operational settings for the system.

The actuators (coils and valves) implement the new settings to control the thruster system. The sensors perform immediate validation of the new parameters to verify system stability. The digital twin model updates its system through new data input which maintains perfect operation between virtual and actual physical systems.

Brief Description of the Drawings :

The operational flowchart in FIG. 1 shows how the adaptive AI-powered thruster system performs its iterative optimization process by using closed-loop systems which connect physical sensor measurements to digital twin simulations and automatic system parameter changes..

Description of Preferred Embodiment :

Figure 1 explains how the adaptive Hall Effect Thruster testbed operates. The system reaches its optimal performance through an ongoing process that continuously connects hardware movement with software simulation.

The system architecture follows a cyclical refinement protocol:

- The process starts at the Start block which enables integrated Sensors to collect real-time physical parameters that include discharge current and magnetic field intensity and propellant flow rate. The AI Controller uses these data points to evaluate the present condition by comparing it with past performance records which enables it to suggest the best operational settings.
- The system uses its Actuators to execute precise adjustments of the thruster settings after it receives AI-generated recommendations. This stage represents the transition from digital command to physical mechanical or electrical change within the test environment..
- Validation and Modeling: The Sensors proceed with their second scan process after completing their first physical adjustment upkeep to confirm that their system parameters now match the expected operational state. The Digital Twin system receives the validated data which it uses to update its internal model according to the new physical conditions of the thruster.
- Optimization Logic: The final stage of the loop involves a decision gate: "Is the Most Efficient Model Acquired?"
- No: If the efficiency threshold or the "Golden Model" target has not been met, the system triggers a "No" branch, returning the logic flow to the AI Controller for a new iteration of set-point recommendations.
- Yes: Once the system determines that the thruster is operating at peak efficiency or that the digital twin accurately mirrors the most optimized physical state, it proceeds to the End state.

Validation and Modeling: The Sensors begin their second scanning process after they finish their first physical maintenance work to determine whether their system performance now matches the expected operational parameters. The Digital Twin system receives the validated data which it uses to update its internal model according to the new physical conditions of the thruster.

Claims

1. ***A performance optimization and autonomous control instrument for Hall-effect thrusters (HET) comprising:***

The system consists of a performance optimization instrument along with an autonomous control system which enables Hall-effect thrusters (HET) to operate. The system includes a power source which supplies primary electrical energy for both plasma generation and logic control operations. The multi-sensor array consists of Hall sensors and pressure gauges and photodiodes which will deliver real-time monitoring data about discharge voltage and current and magnetic field and plume behavior. The central processing unit of the system runs digital twin simulation together with predictive AI model which assesses thruster performance. The system uses actuators which include DC coils and proportional valves to execute thrust control operations that the controller initiates. The system uses a closed-loop feedback system to provide local and remote visual display of both thruster status and system telemetry information.

2. The artificial intelligence controller of the thruster optimization system described in claim 1 employs reinforcement learning through DDPG and PPO to perform automatic data-based tuning without needing human operators.

3. The digital twin system in the thruster optimization system from claim 1 creates an initial simulation validation system which stops unsafe recommendations that would result in unstable plasma oscillations and dangerous equipment conditions.

4. The sensor array of the thruster optimization system from claim 1 uses Kalman filters to eliminate data noise while measuring accuracy during high-voltage testing.

5. The processing unit of the thruster optimization system from claim 1 enables dual- interface monitoring through its ability to send synchronized data to both a local 20x4 LCD display and a remote mobile IoT application.

6. The predictive AI model of the system in claim 1 predicts performance changes to trigger actuator parameter changes before reaching a fixed threshold which helps maintain operational efficiency.

Digitally Signed By:
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Date: 15-06-2026 16:09:10